Nederlandse norm

NEN-EN-ISO 7730

(en)

Klimaatomstandigheden - Analytische bepaling en interpretatie van thermische behaaglijkheid door berekeningen van de PMV en PPD-waarden en lokale thermische behaaglijkheid (ISO 7730:2025,IDT)

Ergonomics of the thermal environment - Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria (ISO 7730:2025,IDT)

Vervangt NEN-EN-ISO 7730:2005

ICS 13.180

oktober 2025

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN ISO 7730

September 2025

ICS 13.180

Supersedes EN ISO 7730:2005

English Version

Ergonomics of the thermal environment - Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria (ISO 7730:2025)

Ergonomie des ambiances thermiques - Détermination analytique et interprétation du confort thermique par le calcul des indices PMV et PPD et par des critères de confort thermique local (ISO 7730:2025)

Ergonomie der thermischen Umgebung - Analytische Bestimmung und Interpretation der thermischen Behaglichkeit durch Berechnung des PMV- und des PPD-Indexes und Kriterien der lokalen thermischen Behaglichkeit (ISO 7730:2025)

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Contents

Page

European foreword **3**

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International Standard

ISO 7730

Ergonomics of the thermal environment — Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria

Ergonomie des ambiances thermiques — Détermination analytique et interprétation du confort thermique par le calcul des indices PMV et PPD et par des critères de confort thermique local

Fourth edition
2025-09

Reference number
ISO 7730:2025(en)

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Contents

Page

Foreword ... **iv**

Introduction ... **v**

1 Scope ... **1**

2 Normative references ... **1**

3 Terms and definitions ... **1**

4 Whole body thermal comfort predicted mean vote (PMV) ... **2**
4.1 Determination ... 2
4.2 Applications ... 4

5 Predicted percentage dissatisfied (PPD) ... **4**

6 Local thermal comfort ... **5**
6.1 General ... 5
6.2 Draught ... 6
6.3 Vertical air temperature difference ... 6
6.4 Warm and cool floors ... 7
6.5 Radiant temperature asymmetry ... 8

7 Thermal environments for comfort ... **9**

8 Non-steady-state thermal environments ... **10**
8.1 General ... 10
8.2 Temperature cycles ... 10
8.3 Temperature drifts or ramps ... 10
8.4 Transients ... 10

Annex A (informative) **Examples of thermal comfort requirements for different categoriesof environment and types of space** ... **11**

Annex B (informative) **Metabolic rates of different activities** ... **16**

Annex C (informative) **Estimation of thermal insulation of clothing ensembles** ... **17**

Annex D (normative) **Computer program for calculating PMV and PPD** ... **21**

Annex E (informative) **Graphics for determination of predicted mean vote (PMV)** ... **25**

Annex F (informative) **Humidity** ... **29**

Annex G (informative) **Air velocity** ... **30**

Bibliography ... **32**

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This document was prepared by Technical Committee ISO/TC 159, *Ergonomics*, Subcommittee SC 5, *Ergonomics of the physical environment*, in collaboration with the European Committee for Standardization (CEN) Technical Committee CEN/TC 122, *Ergonomics*, in accordance with the Agreement on technical cooperation between ISO and CEN (Vienna Agreement).

This fourth edition cancels and replaces the third edition (ISO 7730:2005), which has been technically revised.

The main changes are as follows:

— deletion of sections of the text (long-term evaluations, adaptation and diversity);

— correction of the calculation program;

— deletion of tables for predicting predicted mean vote (PMV).

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Introduction

This document, covering the evaluation of moderate thermal environments, is one of a number of ISO documents (alongside ISO 7243, ISO 7933 and ISO 11079, all dealing with extreme environmental conditions) specifying methods for the measurement and evaluation of the moderate and extreme thermal environments to which human beings are exposed.

A human being's thermal sensation is mainly related to the thermal balance of his or her body as a whole. This balance is influenced by physical activity and clothing, as well as the environmental parameters such as air temperature, mean radiant temperature, air velocity and air humidity. When these factors have been estimated or measured, the index for thermal comfort predicted mean vote (PMV) can be calculated. See Clause 4.

The predicted percentage dissatisfied (PPD) index provides information on thermal discomfort or thermal dissatisfaction expressed as the percentage of people likely to feel too warm or too cool in a given environment. The PPD can be obtained from the PMV. See Clause 5.

Thermal discomfort can also be caused by unwanted local cooling or heating of the body. The most common local discomfort factors are radiant temperature asymmetry (cold or warm surfaces), draught (defined as a local cooling of the body caused by air movement), vertical air temperature difference and cold or warm floors. Clause 6 specifies how to predict the percentage dissatisfied owing to local discomfort parameters.

Dissatisfaction can be caused by hot or cold discomfort for the body as a whole. Comfort limits can, in this case, be expressed by the PMV and PPD indices. But thermal dissatisfaction can also be caused by local thermal discomfort parameters. Clause 7 deals with acceptable thermal environments for comfort.

Clauses 6 and 7 are based mainly on steady-state conditions. Means of evaluating non-steady-state conditions, such as transients (temperature steps), cycling temperatures or temperature ramps, are presented in Clause 8. Thermal environments in buildings or workplaces change over time and it is not always possible to keep conditions within recommended limits.

This document is intended to be used together with ISO/TR 23663. It is also intended to be used along with ISO 28803 when considering persons with special requirements, such as those with physical disabilities. Ethnic, national or geographical differences are also important, especially when considering non-conditioned spaces. Guidance is given in Clause 8 and 10 in ISO/TR 23663.

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International Standard **ISO 7730:2025(en)**

Ergonomics of the thermal environment — Analytical determination and interpretation of thermal comfort using calculation of the PMV and PPD indices and local thermal comfort criteria

1 Scope

This document specifies a method to evaluate the general thermal comfort of people in a space and the degree of discomfort (thermal dissatisfaction) of people exposed to moderate thermal environments. It defines the analytical determination and interpretation of thermal comfort using calculation of predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD) and local thermal comfort criteria, giving the environmental conditions considered acceptable for general thermal comfort as well as those representing local discomfort.

It is applicable to healthy men and women exposed to indoor environments where thermal comfort is desirable, but where moderate deviations from thermal comfort occur, in the design of new environments or the assessment of existing ones.

Although developed specifically for the work environment, this document is applicable to other kinds of environment as well.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 9920, *Ergonomics of the thermal environment — Estimation of thermal insulation and water vapour resistance of a clothing ensemble*

ISO 13731, *Ergonomics of the thermal environment — Vocabulary and symbols*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 13731 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org/

3.1
temperature cycle
variable temperature with a given amplitude and frequency

3.2
temperature drift
passive monotonic, steady, non-cyclic change in the operative temperature of an enclosed space

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3.3
temperature ramp
actively controlled monotonic, steady, non-cyclic change in the operative temperature of an enclosed space

3.4
operative temperature
uniform temperature of an imaginary black enclosure in which an occupant exchanges the same amount of heat by radiation and convection as in the actual non-uniform environment

3.5
transient
sudden change in the thermal conditions due to step change in temperature, humidity, activity or clothing

3.6
draught
unwanted local cooling of the body caused by air movement

4 Whole body thermal comfort predicted mean vote (PMV)

4.1 Determination

The PMV is an index that predicts the mean value of the votes of a large group of persons on the seven-point thermal sensation scale (see Table 1), based on the heat balance of the human body. Thermal balance is obtained when the internal heat production in the body is equal to the loss of heat to the environment. In a moderate environment, the human thermoregulatory system will automatically attempt to modify skin temperature and sweat secretion to maintain heat balance.

Table 1 — Seven-point thermal sensation scale

+ 3	Hot
+ 2	Warm
+ 1	Slightly warm
0	Neutral
− 1	Slightly cool
−2	Cool
− 3	Cold

Calculate the PMV using Formulae (1) to (4):

$$P_{\mathrm{PMV}} = [0{,}303 \cdot \exp(-0{,}036 \cdot M) + 0{,}028] *$$

$$\left\{\begin{array}{l} (M-W) - 3{,}05 \cdot 10^{-3} \cdot [5\,733 - 6{,}99 \cdot (M-W) - p_{\mathrm{a}}] - 0{,}42 \cdot [(M-W) - 58{,}15] \\ -1{,}7 \cdot 10^{-5} \cdot M \cdot (5\,867 - p_{\mathrm{a}}) - 0{,}001\,4 \cdot M \cdot (34 - t_{\mathrm{a}}) \\ -3{,}96 \cdot 10^{-8} \cdot f_{\mathrm{cl}} \cdot \left[(t_{\mathrm{cl}} + 273)^4 - (\bar{t}_{\mathrm{r}} + 273)^4\right] - f_{\mathrm{cl}} \cdot h_{\mathrm{c}} \cdot (t_{\mathrm{cl}} - t_{\mathrm{a}}) \end{array}\right\} \quad (1)$$

$$t_{\mathrm{cl}} = 35{,}7 - 0{,}028 \cdot (M-W) - I_{\mathrm{cl}} \cdot \left\{3{,}96 \cdot 10^{-8} \cdot f_{\mathrm{cl}} \cdot \left[(t_{\mathrm{cl}} + 273)^4 - (\bar{t}_{\mathrm{r}} + 273)^4\right] + f_{\mathrm{cl}} \cdot h_{\mathrm{c}} \cdot (t_{\mathrm{cl}} - t_{\mathrm{a}})\right\} \quad (2)$$

$$h_{\mathrm{c}} = \begin{cases} 2{,}38 \cdot |t_{\mathrm{cl}} - t_{\mathrm{a}}|^{0{,}25} & \text{for} \quad 2{,}38 \cdot |t_{\mathrm{cl}} - t_{\mathrm{a}}|^{0{,}25} > 12{,}1 \cdot \sqrt{v_{\mathrm{ar}}} \\ 12{,}1 \cdot \sqrt{v_{\mathrm{ar}}} & \text{for} \quad 2{,}38 \cdot |t_{\mathrm{cl}} - t_{\mathrm{a}}|^{0{,}25} < 12{,}1 \cdot \sqrt{v_{\mathrm{ar}}} \end{cases} \quad (3)$$

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$$f_{cl} = \begin{cases} 1{,}00 + 1{,}290\, l_{cl} & \text{for } l_{cl} \text{ u } 0{,}078\ \mathrm{m^2 \cdot K/W} \\ 1{,}05 + 0{,}645\, l_{cl} & \text{for } l_{cl} > 0{,}078\ \mathrm{m^2 \cdot K/W} \end{cases} \quad (4)$$

where

P_{PMV} is the PMV;

M is the metabolic rate, in watts per square metre (W/m^2);

W is the effective mechanical power, in watts per square metre (W/m^2);

I_{cl} is the clothing insulation, in square metres kelvin per watt (m^2 · K/W);

f_{cl} is the clothing surface area factor;

t_a is the air temperature, in degrees Celsius (°C);

$\bar{t}_r$ is the mean radiant temperature, in degrees Celsius (°C);

v_{ar} is the relative air velocity, in metres per second (m/s);

p_a is the water vapour partial pressure, in pascals (Pa);

h_c is the convective heat transfer coefficient, in watts per square metre kelvin [W/(m^2 · K)];

t_{cl} is the clothing surface temperature, in degrees Celsius (°C).

NOTE 1 1 metabolic unit = 1 met = 58,2 W/m^2; 1 clothing unit = 1 clo = 0,155 m^2 · °C/W.

PMV can be calculated for different combinations of metabolic rate, clothing insulation, air temperature, mean radiant temperature, air velocity and air humidity. Formulae (2) and (3) for t_{cl} and h_c can be solved by iteration.

The PMV index is derived for steady-state conditions but can be applied with good approximation during minor fluctuations of one or more of the variables, provided that time-weighted averages of the variables during the previous 1 h period are applied.

The index should be used only for values of PMV between −2 and +2, and when the six main parameters are within the following intervals:

M 46 W/m^2 to 232 W/m^2 (0,8 met to 4 met);

I_{cl} 0 m^2 · K/W to 0,310 m^2 · K/W (0 clo to 2 clo);

t_a 10 °C to 30 °C;

$\bar{t}_r$ 10 °C to 40 °C;

v_{ar} 0 m/s to 1 m/s;

p_a 0 Pa to 2 700 Pa.

NOTE 2 In respect of v_{ar}, during light, mainly sedentary, activity, a mean velocity within this range can be felt as a draught.

The metabolic rate to be used in Formulae (1) and (2) can be estimated using values shown in Annex B or in ISO 8996. In the latter case, the value in watts estimated according to ISO 8996 should be divided by the commonly accepted mean value of the body surface area, ADu = 1,8 m^2. For varying metabolic rates, a time-weighted average should be estimated during the previous 1 h period. Estimate the thermal resistance of clothing and chair using ISO 9920 or Annex C, taking into account the season of year.

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Determine the PMV in one of the following ways:

a) From Formula (1) using a digital computer. A BASIC program is given in Annex D for this purpose. For verification of other computer programs, Annex D provides example output.

b) From Annex E, where graphics of PMV values are given for different combinations of activity, clothing, operative temperature and relative velocity.

c) By measurement, using an integrating sensor (equivalent and operative temperatures).

The influence of humidity on thermal sensation is small at moderate temperatures close to comfort and can usually be disregarded when determining the PMV value (see Annex F).

4.2 Applications

The PMV can be used to check whether a given thermal environment conforms to comfort criteria (see Clause 7 and Annex A) and to establish requirements for different categories of acceptability.

By setting PMV = 0, an equation is established which predicts combinations of activity, clothing and environmental parameters which on average will provide a thermally neutral sensation.

5 Predicted percentage dissatisfied (PPD)

The PMV index is defined in relation to the mean value of the thermal votes of a large group of people exposed to the same environment. But individual votes are scattered around this mean value and it is useful to be able to predict the number of people likely to feel uncomfortably warm or cool.

The PPD is an index that establishes a quantitative index related to the percentage of thermally dissatisfied people who feel too cool or too warm. For the purposes of this document, thermally dissatisfied people are those who will vote 'hot', 'warm', 'cool' or 'cold' on the seven-point thermal sensation scale given in Table 1.

With the PMV value determined, calculate the PPD using Formula (5), see Figure 1:

$$P_{\mathrm{PPD}} = 100 - 95 \cdot \exp\left(-0{,}033\,53 \cdot P_{\mathrm{PMV}}{}^{4} - 0{,}217\,9 \cdot P_{\mathrm{PMV}}{}^{2}\right) \tag{5}$$

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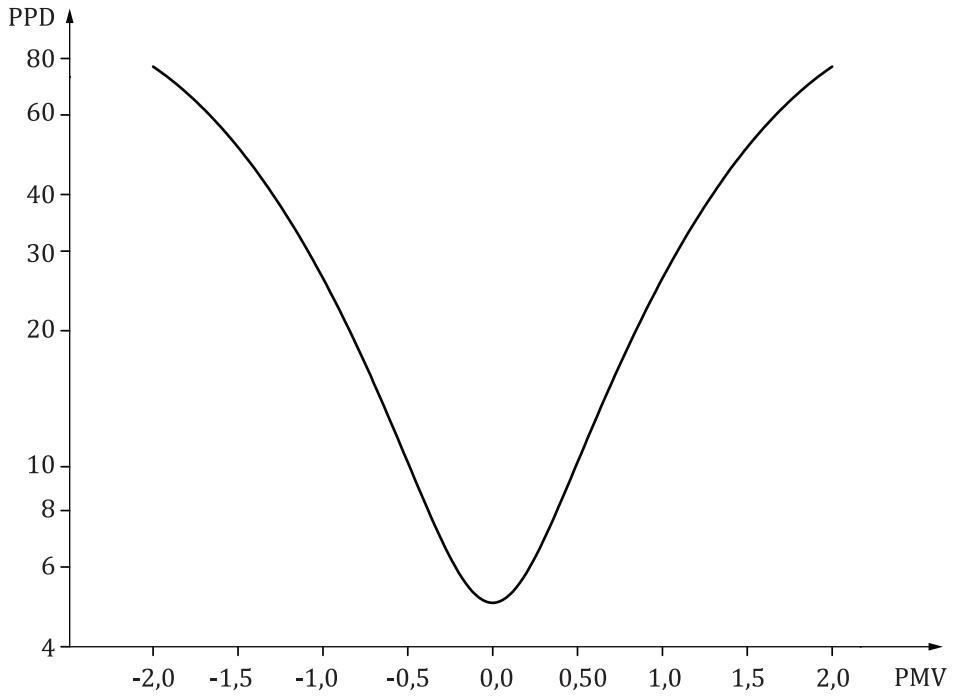

Key

PMV is the PMV;

PPD is the PPD, %

Figure 1 — PPD as a function of PMV

The PPD is related to the number of thermally dissatisfied persons among a large group of people. The rest of the group will feel thermally neutral, slightly warm or slightly cool. The relation between PMV and PPD distribution is given in Table 2.

Table 2 — Distribution of individual thermal sensation votes for different values of mean vote

PMV	PPD	Persons predicted to vote [a] %		
		0	−1, 0 or +1	−2, −1, 0, +1 or +2
+2	75	5	25	70
+1	25	30	75	95
+0,5	10	55	90	98
0	5	60	95	100
−0,5	10	55	90	98
−1	25	30	75	95
−2	75	5	25	70

[a] Based on experiments involving 1 300 subjects.

6 Local thermal comfort

6.1 General

The PMV and PPD are indices related to warm and cold discomfort for the body as a whole. But thermal dissatisfaction can also be caused by unwanted cold or warm sensation of a particular part of the body. This is known as local discomfort. The most common cause of local discomfort is draught (6.2). But local discomfort can also be caused by an abnormally high vertical temperature difference between the head and

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ankles (6.3), by too warm or too cool a floor (6.4) or by a radiant temperature asymmetry (6.5). Annex A provides examples of local and overall thermal comfort requirements for different categories of environment and types of space.

It is mainly people at light sedentary activity who are sensitive to local discomfort. These will have a thermal sensation for the whole body close to neutral. At higher levels of activity, people are less thermally sensitive and, consequently, the risk of local discomfort is lower.

6.2 Draught

The discomfort due to draught can be expressed as the percentage of people predicted to be bothered by draught. Calculate the draught rate (*DR*) using Formula (6) (model of draught):

$$P_{\mathrm{DR}} = \left(34 - t_{\mathrm{a,l}}\right)\left(\bar{v}_{\mathrm{a,l}} - 0{,}05\right)^{0{,}62}\left(0{,}37 \cdot \bar{v}_{\mathrm{a,l}} \cdot T_{\mathrm{u}} + 3{,}14\right) \tag{6}$$

where

P_{DR} is the DR;

$t_{\mathrm{a,l}}$ is the local air temperature, in degrees Celsius, 20 °C to 26 °C;

$\bar{v}_{\mathrm{a,l}}$ is the local mean air velocity, in metres per second, < 0,5 m/s;

T_{u} is the local turbulence intensity, in percent, 10 % to 60 % (if unknown, 40 % may be used).

For < 0,05 m/s, use = 0,05 m/s.

For P_{DR} > 100 %, use P_{DR} = 100 %.

The model applies to people at light, mainly sedentary activity with a thermal sensation for the whole body close to neutral and for prediction of draught at the neck. At the level of arms and feet, the model can overestimate the predicted draught rate. The sensation of draught is lower at activities higher than sedentary (> 1,2 met) and for people feeling warmer than neutral. Additional information on the effect of air velocity can be found in Annex G.

6.3 Vertical air temperature difference

A high vertical air temperature difference between head and ankles can cause discomfort. Figure 2 shows the percentage dissatisfied (PD) as a function of the vertical air temperature difference between head and ankles. The figure applies when the temperature increases upwards. People are less sensitive under decreasing temperatures. Determine the PD using Formula (7):

$$P_{\mathrm{PD}} = \frac{100}{1 + \exp\left(5{,}76 - 0{,}856 \cdot \Delta t_{\mathrm{a,v}}\right)} \tag{7}$$

Formula (7), derived from the original data using logistic regression analysis, should only be used at $\Delta t_{\mathrm{a,v}}$ < 8 °C.

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Key

PD percentage dissatisfied, %

$\Delta t_{a,v}$ vertical air temperature difference between head and feet, °C

Figure 2 — Local discomfort caused by vertical air temperature difference

6.4 Warm and cool floors

If the floor is too warm or too cool, the occupants of the space can feel uncomfortable owing to thermal sensation of their feet. For people wearing light indoor shoes, it is the temperature of the floor rather than the material of the floor covering which is important for comfort. Figure 3 shows the percentage dissatisfied as a function of the floor temperature, based on studies with either standing or sedentary people, or both.

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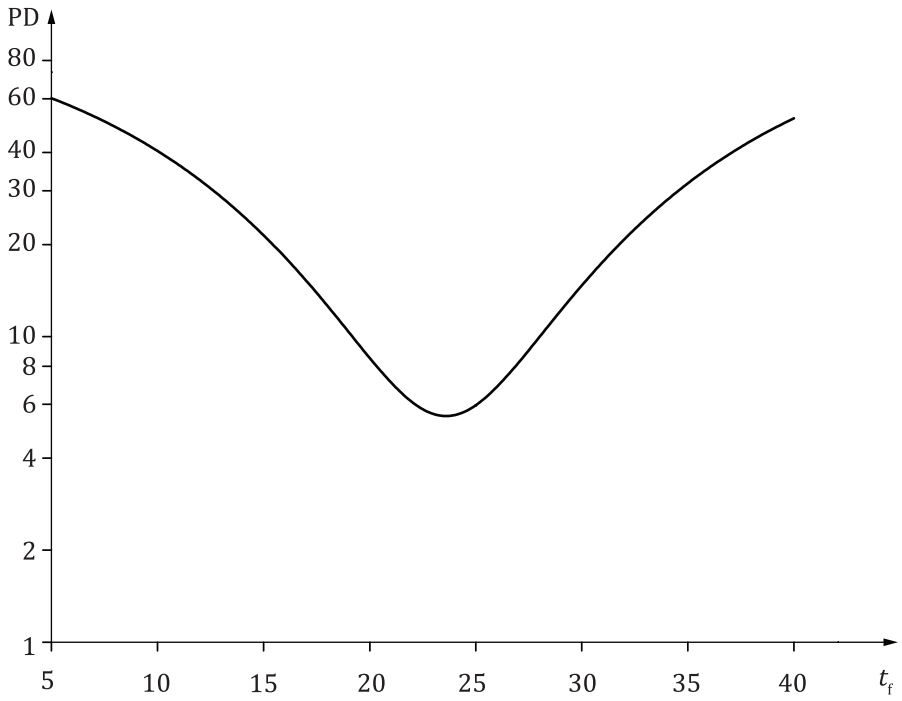

Key

PD percentage dissatisfied, %

t_f floor temperature, °C

Figure 3 — Local thermal discomfort caused by warm or cold floors

For people sitting or lying on the floor, similar values may be used. Determine the PD using Formula (8), derived from the original data using non-linear regression analysis:

$$P_{PD} = 100 - 94 \cdot \exp(-1{,}387 + 0{,}118 \cdot t_f - 0{,}0025 \cdot t_f^2) \quad (8)$$

For longer occupancy, the results are not valid for electrically heated floors.

NOTE By electrical heating, a certain heat input is provided independent of the surface temperature. A water-based heating system will not produce temperatures higher than the water temperature.

For spaces that people occupy with bare feet, see ISO/TS 13732-2.

6.5 Radiant temperature asymmetry

Radiant temperature asymmetry (Δt_{pr}) can also cause discomfort. People are most sensitive to radiant asymmetry caused by warm ceilings or cool walls (windows). Figure 4 shows the percentage dissatisfied as a function of the radiant temperature asymmetry caused by a warm ceiling, cool wall, cool ceiling or warm wall. For horizontal radiant asymmetry, Figure 4 applies from side-to-side (left–right or right–left) asymmetry, the curves providing a conservative estimate of the discomfort: no other positions of the body in relation to the surfaces (e.g. front–back) cause higher asymmetry discomfort. Determine the PD using Formula (9) as applicable.

$$P_{PD} = 100/(1+\exp(K_1 - K_2 * \Delta t_{pr})) - K_3 \quad (9)$$

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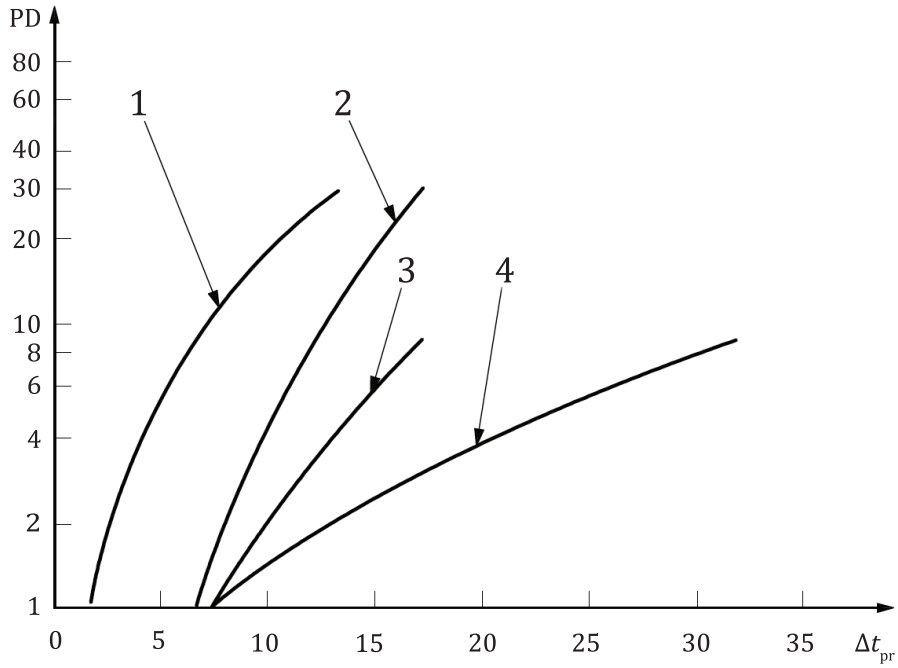
Table 3 — Constants used in Formula (9) for different types of radiant asymmetry

Asymmetry	Δt_{pr}limit	K_1	K_2	K_3
1: warm ceiling	< 23 K	2,94	0,166	5,5
2: cool wall	< 15 K	5,89	0,297	1
3: cool ceiling	< 15 K	5,19	0,173	1
4: warm wall	< 35 K	3,41	0,044	3,5

The numbers refer to the corresponding curves in Figure 4.

Formula (9) was derived from the original data using logistic regression analysis and should not be used beyond the ranges shown in Table 3. Those for 1) (warm ceiling) and for 4) (warm wall) have been adjusted to account for discomfort not caused by radiant asymmetry. See Figure 4.



Key

PD percentage dissatisfied, %

Δt_{pr} radiant temperature asymmetry, °C

1 warm ceiling

2 cool wall

3 cool ceiling

4 warm wall

Figure 4 — Local thermal discomfort caused by radiant temperature asymmetry

7 Thermal environments for comfort

Thermal comfort is that condition of mind which expresses satisfaction with the thermal environment. Dissatisfaction can be caused by warm or cool discomfort of the body as a whole, as expressed by the PMV and PPD, or by a warm or cold sensation of one particular part of the body (local comfort).

Due to individual differences, it is impossible to specify a thermal environment that will satisfy everybody. There will always be a percentage of dissatisfied occupants.

Often it will be the same persons who are sensitive to different types of local discomfort. For instance, a person sensitive to draught can also be sensitive to local cooling caused by radiant asymmetry or by a cold

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floor. Such a cold-sensitive person can also more easily experience cool discomfort for the body as a whole. Therefore, the PPD, DR or PD caused by other types of local discomfort should not be added.

Due to local or national priorities, technical developments and climatic regions, a higher thermal quality (fewer dissatisfied) or lower quality (more dissatisfied) in some cases may be accepted. In such cases, the PMV and PPD, the model of draught, the relation between local thermal discomfort parameters (see Clause 6) and the expected percentage of dissatisfied people may be used to determine different ranges of environmental parameters for the evaluation and design of the thermal environment.

Examples of different categories of requirements are given in Annex A.

8 Non-steady-state thermal environments

8.1 General

The basis for the methods given in Clauses 4 to 6 is steady-state conditions. The thermal environment is, however, often in a non-steady-state and the question arises as to whether the methods then apply. Three types of non-steady-state conditions can occur: temperature cycles, temperature drifts or ramps and transients.

8.2 Temperature cycles

Temperature cycles can occur due to the control of the temperature in a space. If the peak-to-peak variation is less than 1 K, there will be no influence on the comfort and the recommendations for steady-state may be used. Higher peak variations can decrease comfort.

8.3 Temperature drifts or ramps

If the rate of temperature change for drifts or ramps is lower than 4,0 K/h, the methods for steady-state variation apply.

8.4 Transients

In general, the following statements regarding transients can be made:

— A step-change of operative temperature is felt instantaneously.

— After an up-step in operative temperature, the new steady-state thermal sensation is experienced immediately, i.e. the PMV-PPD indices can be used.

— Following a down-step in operative temperature, the thermal sensation drops at first to a level beneath the PMV index, then increases and reaches under steady-state conditions the steady-state level after approximately 30 min, i.e. the PMV-PPD index show values that are too high for the first 30 min. The time to reach a new steady-state condition depends on the initial conditions.

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Annex A
(informative)

Examples of thermal comfort requirements for different categoriesof environment and types of space

A.1 Categories of thermal environment

The desired thermal environment for a space may be selected from among the four categories, I, II, III and IV, according to Table A.1. For local discomfort, only three categories apply. All the criteria should be satisfied simultaneously for each category.

Table A.1 — Categories of thermal environment

Category	Thermal state of the body as a whole		Local discomfort			
	PPD %	PMV	DR %	PD %		
				vertical air temperature difference	caused by warm or cool floor	radiant asym-metry
I	< 6	− 0,2 < PMV < + 0,2	< 10	< 3	< 10	< 5
II	< 10	− 0,5 < PMV < + 0,5	< 20	< 5	< 10	< 5
III	< 15	− 0,7 < PMV < + 0,7	< 30	< 10	< 15	< 10
IV	< 25	− 1,0 < PMV < + 1,0				

Each category prescribes a maximum percentage dissatisfied for the body as a whole (PPD) and a PD for each of the four types of local discomfort. Some requirements are difficult to meet in practice while others are quite easily met. The different percentages express a balance struck between the aim of a few dissatisfied persons and what is practically obtainable using existing technology.

Owing to the accuracy of instrumentation for measuring the input parameters according to ISO 7726, it can be difficult to verify that the PMV conforms to the class A category (−0,2 < PMV < +0,2). Instead, the verification may be based on the corresponding operative temperature range, as specified in A.2 and Table A.5.

The four categories presented in Table A.1 apply to spaces where persons are exposed to the same thermal environment. It is an advantage if some kind of individual control of the thermal environment can be established for each person in a space. Individual control of the local air temperature, mean radiant temperature or air velocity can contribute to balancing the rather large differences between individual requirements and consequently can lead to fewer dissatisfied persons.

Modification of clothing can also contribute to balance individual differences. The effect on the optimum operative temperature of adding or removing different garments is described in Table C.2.

A.2 Operative temperature range

For a given space there exists an optimum operative temperature corresponding to PMV = 0, depending on the activity and the clothing of the occupants. Figure A.1 shows the optimum operative temperature and the permissible temperature range as a function of clothing and activity for three of the four categories. The optimum operative temperature is the same for the three categories, while the permissible range around the optimum operative temperature varies.

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The operative temperature at all locations within the occupied zone of a space should, at all times, be within the permissible range. This means that the permissible range should cover both spatial and temporal variations, including fluctuations caused by the control system.

Figure A.1 applies for a relative humidity of 50 %; however, in moderate environments the air humidity has only a modest impact on the thermal sensation. Typically, a 10 % higher relative humidity and a 0,3 °C higher operative temperature are perceived as being warmer in equal measure.

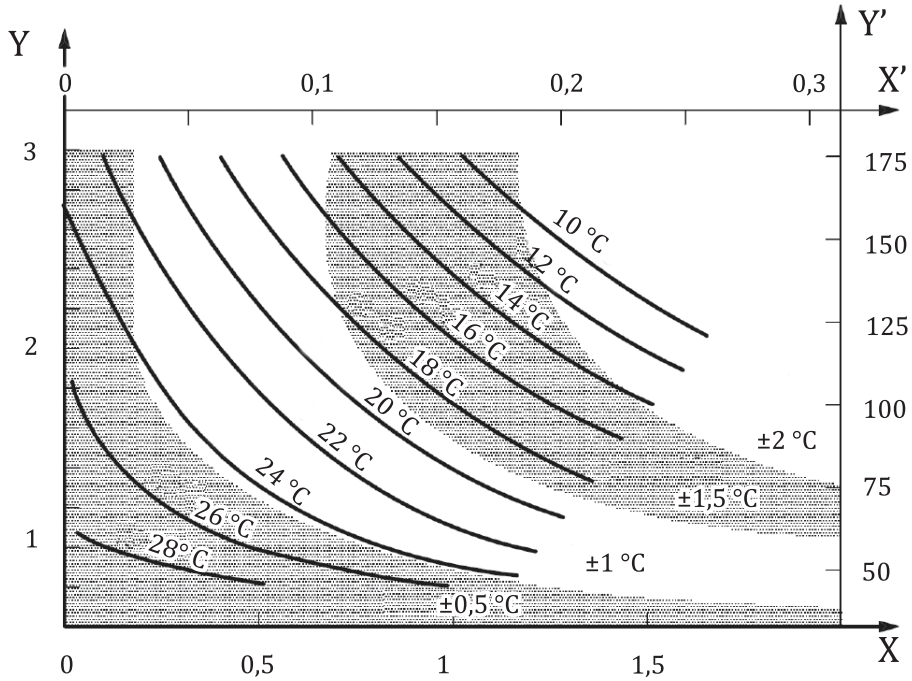
The PDs in Table A.1 are not additive. In practice, a higher or lower number of dissatisfied persons can be found when using subjective questionnaires in field investigations (see ISO 10551).

The air velocity in the space is assumed to be < 0,1 m/s. The relative air velocity, v_{ar}, caused by body movement is estimated to be zero for a metabolic rate, M, less than 1 met and $v_{ar} = 0{,}3\ (M - 1)$ for $M > 1$ met. The diagrams are determined for a relative humidity = 50 %, but the humidity only has a slight influence on the optimum and permissible temperature ranges.

A.3 Local thermal comfort

Figure A.2 give ranges for local thermal comfort due to draught for the three categories presented in Table A.1.

The maximum allowable mean air velocity is a function of local air temperature and turbulence intensity. The turbulence intensity can vary between 30 % and 60 % in spaces with mixed-flow air distribution. In spaces with displacement ventilation or without mechanical ventilation, the turbulence intensity can be lower.



a) Category I: PPD < 6 %

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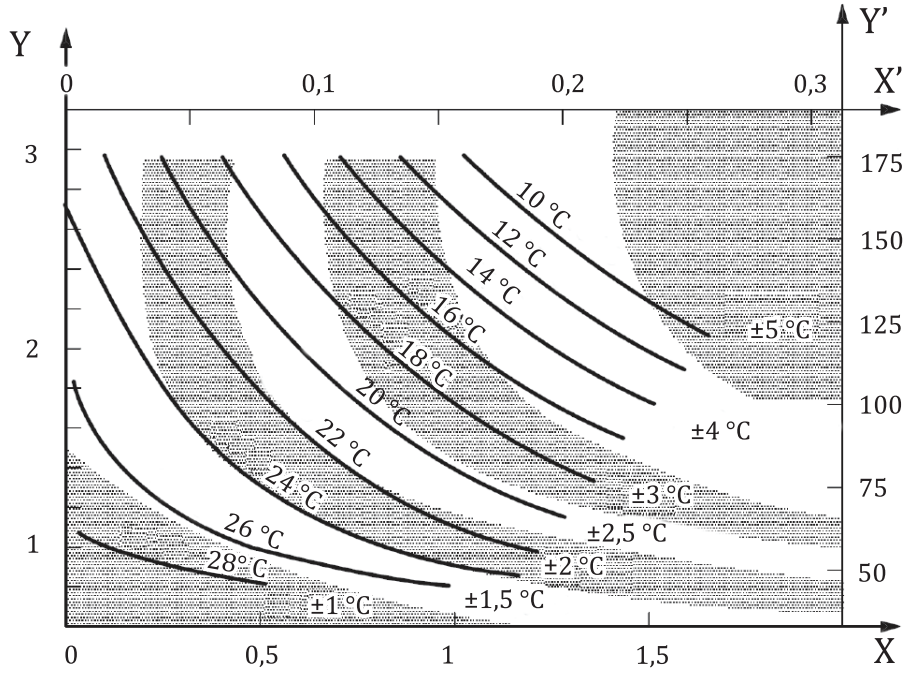
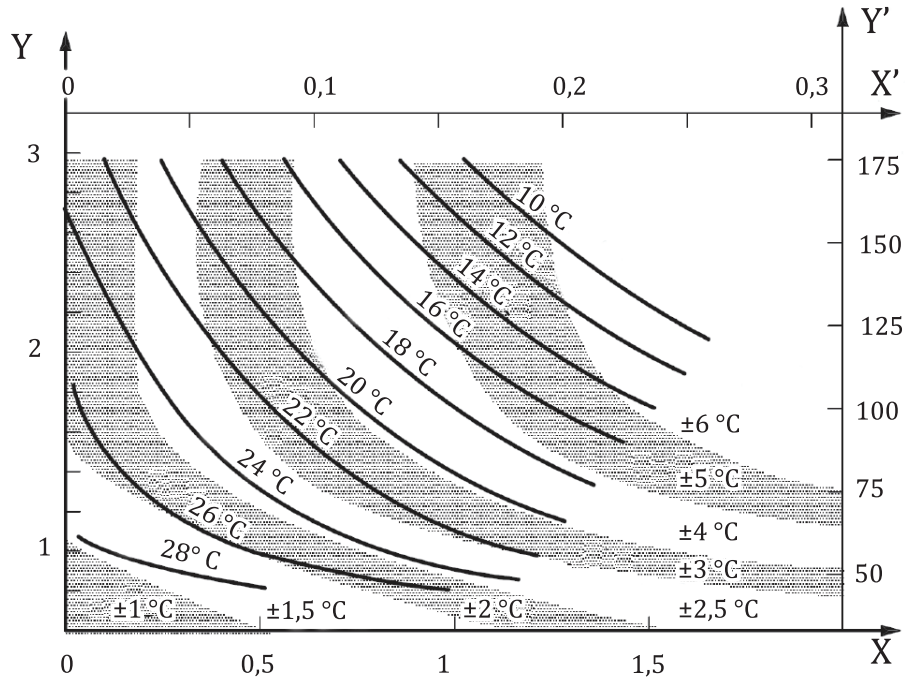


b) Category II: PPD < 10 %



c) Category III: PPD < 15 %

Key

X basic clothing insulation, in clothing units, (clo)

X' basic clothing insulation, in clothing units, $m^2 \cdot °C/W$

Y metabolic rate, in metabolic units, (met)

Y' metabolic rate, in metabolic units, W/m^2

PPD predicted percentage dissatisfied, %

NOTE The diagrams also show the range around the optimum temperature for the three categories.

Figure A.1 — Optimal operative temperature and the permissible temperature range

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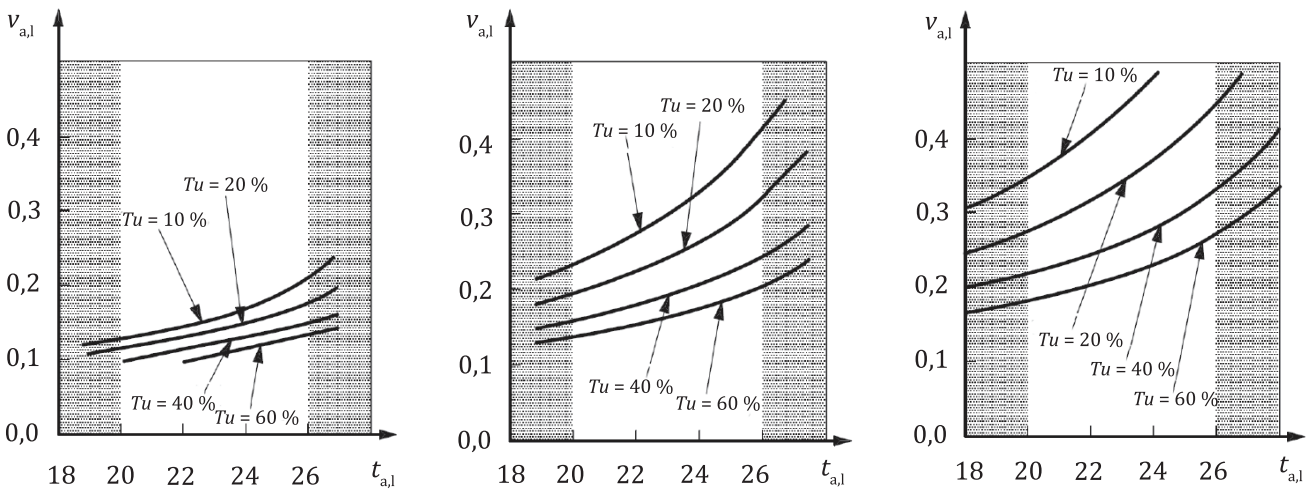

a) Category I: DR = 10 % **b) Category II: DR = 20 %** **c) Category III: DR = 30 %**

Key

$t_{a,l}$ local air temperature, °C

$\bar{v}_{a,l}$ local mean air velocity, m/s

Tu turbulence intensity, %

Figure A.2 — Maximum allowable mean air velocity as function of local air temperature and turbulence intensity

Tables A.2, A.3 and A.4 give values for-local thermal comfort related to vertical air temperature difference, warm or cold floor and radiant temperature asymmetry, respectively.

Table A.2 — Vertical air temperature difference between head and ankles

Category	Vertical air temperature difference [a] °C
I	< 2
II	< 3
III	< 4
[a] 1,1 m and 0,1 m above floor.	

Table A.3 — Range of floor temperature

Category	Floor surface temperature range °C
I	19 to 29
II	19 to 29
III	17 to 31

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Table A.4 — Radiant temperature asymmetry

Category	Radiant temperature asymmetry °C			
	Warm ceiling	Cool wall	Cool ceiling	Warm wall
I	< 5	< 10	< 14	< 23
II	< 5	< 10	< 14	< 23
III	< 7	< 13	< 18	< 35

A.4 Design criteria for different types of space — Examples

The design criteria specified in Table A.5 are derived under certain assumptions. For the thermal environment, the criteria for the operative temperature are based on typical levels of activity, for clothing of 0,5 clo during summer ("cooling season") and 1,0 clo during winter ("heating season"). The criteria for the mean air velocity apply for a turbulence intensity of approximately 40 % (mixing ventilation). The design criteria are valid for the occupancy conditions as given but can also be applicable to other types of spaces used in similar ways.

Table A.5 — Example design criteria for spaces in various types of building

Type of building or space	Activity met W/m²	Category	Operative temperature °C		Maximum mean air velocity [a] m/s	
			Summer (cooling season)	Winter (heating season)	Summer (cooling season)	Winter (heating season)
Single office Landscape office Conference room Auditorium Cafeteria or restaurant Classroom	1,2 met 70 W/m²	I	24,5 ± 1,0	22,0 ± 1,0	0,12	0,10
		II	24,5 ± 1,5	22,0 ± 2,0	0,19	0,16
		III	24,5 ± 2,5	22,0 ± 3,0	0,24	0,21 [b]
		IV	24,5 ± 3,5	22,0 ± 4,0		
Kindergarten Standing or sedentary	1,4 met 81 W/m²	I	23,5 ± 1,5	20,5 ± 2,0	0,11	0,10 [b]
		II	23,5 ± 2,0	20,5 ± 3,0	0,18	0,15 [b]
		III	23,5 ± 3,0	20,5 ± 4,0	0,23	0,19 [b]
		IV	23,5 ± 4,0	20,5 ± 5,0		
Department store Standing or walking	1,6 met 93 W/m²	I	23,0 ± 1,5	20,0 ± 2,0	0,16	0,13 [b]
		II	23,0 ± 2,0	20,0 ± 3,0	0,20	0,15 [b]
		III	23,0 ± 3,0	20,0 ± 4,0	0,23	0,18 [b]
		IV	23,0 ± 4,0	20,0 ± 5,0		

[a] The maximum mean air velocity is based on a turbulence intensity of 40 % and air temperature equal to the operative temperature according to 6.2 and Figure A.2. A relative humidity of 60 % and 40 % is used for summer and winter, respectively. For both summer and winter, a lower temperature in the range is used to determine the maximum mean air velocity.

[b] Below 20 °C limit (see Figure A.2).

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Annex B
(informative)

Metabolic rates of different activities

Table B.1 gives examples of the metabolic rate for different activities. Further information on metabolic rates is given in ISO 8996. That elderly people often have a lower average activity than younger people also must be taken into account.

Table B.1 — Examples of metabolic rates

Activity	Metabolic rate	
	W/m^2	met
Reclining	46	0,8
Seated, relaxed	58	1,0
Sedentary activity (office, dwelling, school, laboratory)	70	1,2
Standing, light activity (shopping, laboratory, light industry)	93	1,6
Standing, medium activity (shop assistant, domestic work, machine work)	116	2,0
Walking on level ground:		
2 km/h	110	1,9
3 km/h	140	2,4
4 km/h	165	2,8
5 km/h	200	3,4

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Annex C
(informative)

Estimation of thermal insulation of clothing ensembles

C.1 General

The clothing insulation (I_{cl}) can be estimated directly from the data presented in Table C.1 for typical combinations of garments (the values are for static thermal insulation) or indirectly by summation of the partial insulation values for each item of clothing, I_{clu} , as presented in Table C.2.

Table C.2 gives the corresponding change in the optimum operative temperature necessary to maintain thermal sensation at neutral when a garment is added or removed at light mainly sedentary activity (1,2 met).

For sedentary persons, the chair can contribute an additional insulation of 0 clo to 0,4 clo (see Table C.3). Further information is given in ISO 9920.

Table C.1 — Thermal insulation for typical combinations of garments

Work clothing	I_{cl}		Daily wear clothing	I_{cl}	
	clo	$m^2 \cdot K/W$		clo	$m^2 \cdot K/W$
Underpants, boiler suit, socks, shoes	0,70	0,110	Panties, T-shirt, shorts, light socks, sandals	0,30	0,050
Underpants, shirt, boiler suit, socks, shoes	0,80	0,125	Underpants, shirt with short sleeves, light trousers, light socks, shoes	0,50	0,080
Underpants, shirt, trousers, smock, socks, shoes	0,90	0,140	Panties, petticoat, stockings, dress, shoes	0,70	0,105
Underwear with short sleeves and legs, shirt, trousers, jacket, socks, shoes	1,00	0,155	Underwear, shirt, trousers, socks, shoes	0,70	0,110
Underwear with long legs and sleeves, thermo-jacket, socks, shoes	1,20	0,185	Panties, shirt, trousers, jacket, socks, shoes	1,00	0,155
Underwear with short sleeves and legs, shirt, trousers, jacket, heavy quilted outer jacket and overalls, socks, shoes, cap, gloves	1,40	0,220	Panties, stockings, blouse, long skirt, jacket, shoes	1,10	0,170
Underwear with short sleeves and legs, shirt, trousers, jacket, heavy quilted outer jacket and overalls, socks, shoes	2,00	0,310	Underwear with long sleeves and legs, shirt, trousers, V-neck sweater, jacket, socks, shoes	1,30	0,200
Underwear with long sleeves and legs, thermo-jacket and trousers, parka with heavy quitting, overalls with heavy quilting, socks, shoes, cap, gloves	2,55	0,395	Underwear with short sleeves and legs, shirt, trousers, vest, jacket, coat, socks, shoes	1,50	0,230

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Table C.2 — Thermal insulation for garments and changes of optimum operative temperature

Garment	I_{clu} clo	I_{clu} $m^2 \cdot K/W$	Change of optimum operative temperature °C
Underwear			
Panties	0,03	0,005	0,2
Underpants with long legs	0,10	0,016	0,6
Singlet	0,04	0,006	0,3
T-shirt	0,09	0,014	0,6
Shirt with long sleeves	0,12	0,019	0,8
Panties and bra	0,03	0,005	0,2
Shirts or blouses			
Short sleeves	0,15	0,023	0,9
Light-weight, long sleeves	0,20	0,031	1,3
Normal, long sleeves	0,25	0,039	1,6
Flannel shirt, long sleeves	0,30	0,047	1,9
Lightweight blouse, long sleeves	0,15	0,023	0,9
Trousers			
Shorts	0,06	0,009	0,4
Lightweight	0,20	0,031	1,3
Normal	0,25	0,039	1,6
Flannel	0,28	0,043	1,7
Dresses or skirts			
Light skirts (summer)	0,15	0,023	0,9
Heavy skirt (winter)	0,25	0,039	1,6
Light dress, short sleeves	0,20	0,031	1,3
Winter dress, long sleeves	0,40	0,062	2,5
Boiler suit	0,55	0,085	3,4
Sweaters			
Sleeveless vest	0,12	0,019	0,8
Thin sweater	0,20	0,031	1,3
Sweater	0,28	0,043	1,7
Thick sweater	0,35	0,054	2,2
Jackets			
Light summer jacket	0,25	0,039	1,6
Jacket	0,35	0,054	2,2
Smock	0,30	0,047	1,9
High insulation, fibre-pelt			
Boiler suit	0,90	0,140	5,6
Trousers	0,35	0,054	2,2
Jacket	0,40	0,062	2,5
Vest	0,20	0,031	1,3
Outdoor clothing			
Coat	0,60	0,093	3,7
Down jacket	0,55	0,085	3,4
Parka	0,70	0,109	4,3
Fibre-pelt overalls	0,55	0,085	3,4
Sundries			
Socks	0,02	0,003	0,1
Thick, ankle socks	0,05	0,008	0,3
Thick, long socks	0,10	0,016	0,6
Nylon stockings	0,03	0,005	0,2
Shoes (thin soled)	0,02	0,003	0,1
Shoes (thick soled)	0,04	0,006	0,3
Boots	0,10	0,016	0,6
Gloves	0,05	0,008	0,3

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Table C.3 — Thermal insulation values for chairs

Type of chair	I_{clu}	
	clo	$m^2 \cdot K/W$
Net or metal chair	0,00	0,00
Wooden stool	0,01	0,002
Standard office chair	0,1	0,016
Executive chair	0,15	0,023

The values given in Table C.3 may be added to individual garment insulation values taken from Table C.2 or to the ensemble values from Table C.1.

C.2 Determination of dynamic insulation characteristics of clothing (ISO 9920)

The activity as well as ventilation modify the insulation characteristics of the clothing and the adjacent air layer. Both wind and body movement reduce the insulation, which therefore must be corrected. The correction factor for the static total clothing insulation and the external air layer insulation can be estimated using Formulae (C.1) to (C.3).

For a clothed person in normal or light clothing ($0,6 < I_{cl} < 1,4$ clo or $1,2 < I_T < 2,0$ clo):

$$I_{T,r} = I_T \cdot \mathrm{Corr}, I_T = I_T \cdot \exp\left[-0,281 \cdot (v_{ar} - 0,15) + 0,44 \cdot (v_{ar} - 0,15)^2 - 0,492 \cdot v_w + 0,176 \cdot v_w^2\right] \quad \text{(C.1)}$$

where

$I_{T,r}$ is the resultant total clothing insulation, in square metres kelvin per watt or clothing units ($m^2 \cdot K/W$ or clo);

I_T is the total clothing insulation, in square metres kelvin per watt or clothing units ($m^2 \cdot K/W$ or clo);

Corr, I_T is the correction factor for the total clothing insulation; if $\mathrm{Corr}, I_T > 1$ and/or $\mathrm{Corr}, I_a > 1$ then set their value to 1.

$v_{ar} = v_a + v_w$

v_{ar} is the air velocity relative to the person, in metres per second;

v_w is the walking speed, in metres per second.

For a nude person ($I_{cl} = 0$ clo):

$$I_{a,r} = I_a \cdot \mathrm{Corr}, I_a = I_a \cdot \exp\left[-0,533 \cdot (v_{ar} - 0,15) + 0,069 \cdot (v_{ar} - 0,15)^2 - 0,462 \cdot v_w + 0,201 \cdot v_w^2\right] \quad \text{(C.2)}$$

where

$I_{a,r}$ is the resultant insulation provided by the boundary air layer under the current conditions, in square metres kelvin per watt or clothing units ($m^2 \cdot K/W$ or clo);

I_a is the insulation provided by the boundary air layer, in square metres kelvin per watt or clothing units ($m^2 \cdot K/W$ or clo); $I_a = 0,111$ m^2K/W (see ISO 7933);

Corr, I_a is the correction factor for I_a; if $\mathrm{Corr}, I_T > 1$ and/or $\mathrm{Corr}, I_a > 1$ then set their value to 1.

The resultant dynamic clothing insulation is determined by:

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$$I_{cl,r} = I_{T,r} - \frac{I_{a,r}}{f_{cl}} \quad \text{(C.3)}$$

where

f_{cl} is the clothing area factor; the ratio of the surface area of the clothed body to the surface area of the nude body;

v_{ar} limited to 3,5 m/s;

v_{w} limited to 1,2 m/s.

When the walking speed is undefined or the person is stationary, the value for v_w can be calculated as:

$$v_w = 0{,}005\,2(M - 58) \quad \text{with } v_w \text{ u } 0{,}7\ \text{m/s}$$

For very low clothing insulations, for example for 0 clo ≤ I_{cl} ≤ 0,6 clo, an interpolation between Formulae (C.1) and (C.2) has been derived, see Formula (C.3):

$$\frac{I_{T,r}}{I_T} = \frac{\left((0{,}6 - I_{cl}) \cdot I_{T,r,nude} + I_{cl} \cdot I_{T,r,dressed}\right)}{0{,}6} \quad \text{(C.4)}$$

where

$I_{T,r,dressed}$ is equal to $I_{T,r}$, as determined by Formula (C.1), in clothing units (clo);

$I_{T,r,nude}$ is equal to $I_{a,r}$ determined by Formula (C.2), in clothing units (clo).

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Annex D
(normative)

Computer program for calculating PMV and PPD

The following BASIC program computes the PMV and the PPD for a given set of input variables. Other programming languages may also be used, but the output should be verified with the given BASIC code or with the example values given in Table D.1.

Dynamic effects on clothing insulation shall be corrected before inserting the resultant thermal insulation for the clothing ($I_{cl,r}$) in this equation according to Annex C or ISO 9920.

Variables	Symbols in program
Clothing insulation, clo	CLO
Metabolic rate, met	MET
External work, met	WME
Air temperature, °C	TA
Mean radiant temperature, °C	TR
Relative air velocity, m/s	VEL
Relative humidity, %	RH
Partial water vapour pressure, Pa	PA

```
10   Computer program (BASIC) for calculation of
20   Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD)
30   in accordance with International Standard, ISO 7730
40   CLS:   PRINT   "DATA ENTRY"                                  ; data entry
50   INPUT  "       Clothing insulation                           (clo) ";     CLO
60   INPUT  "       Metabolic rate                                (met) ";     MET
70   INPUT  "       External work, normally around 0              (met) ";     WME
80   INPUT  "       Air temperature                               (°C) ";      TA
90   INPUT  "       Mean radiant temperature                      (°C) ";      TR
100  INPUT  "       Relative air velocity                         (m/s) ";     VEL
110  INPUT  "       ENTER EITHER RH OR WATER VAPOUR PRESSURE BUT NOT BOTH"
120  INPUT  "       Relative humidity                             (%) ";       RH
130  INPUT  "       Water vapour pressure                         (Pa) ";      PA
140  DEF FNPS (T) = EXP (16.6536-4030.183/(T+235))      : saturated vapour pressure, kPa
150  IF PA = 0 THEN PA = RH * 10 * FNPS (TA)            : water vapour pressure, Pa
160  ICL = .155 * CLO                                   : thermal insulation of the clothing in m²K/W
```

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```
170  M = MET * 58.15                                        : metabolic rate in W/m2
180  W = WME * 58.15                                        : external work in W/m2
190  MW = M – W                                             : internal heat production in the human body
200  IF ICL <= 0,078 THEN FCL = 1 + 1.29 * ICL
     ELSE FCL = 1.05 + 0.645 * ICL                          : clothing area factor
210  HCF = 12.1 * SQR (VEL)                                 : heat transf. coeff. by forced convection
220  TAA = TA + 273                                         : air temperature in Kelvin
230  TRA = TR + 273                                         : mean radiant temperature in Kelvin
240  -----CALCULATE SURFACE TEMPERATURE OF CLOTHING BY ITERATION ---
250  TCLA = TAA + (35.5-TA) / (3.5 *(6.45* ICL + .1))       : first guess for surface temperature
                                                            of clothing
260  P1 = ICL * FCL                                         : calculation term
270  P2 = P1 * 3.96                                         : calculation term
280  P3 = P1 * 100                                          : calculation term
290  P4 = P1 * TAA                                          : calculation term
300  P5 = 308.7 - .028 * MW + P2 * (TRA/100) ^ 4
310  XN = TCLA / 100
320  XF = XN
330  N = 0                                                  : N: number of iterations
340  EPS = .00015                                           : stop criteria in iteration
350  XF = (XF + XN)/2
360  HCN =2.38 * ABS (100 * XF – TAA) ^ .25: heat transf. coeff. by natural convection
370  IF HCF>HCN THEN HC = HCF ELSE HC = HCN
380  XN = (P5 + P4 * HC – P2 * XF ^ 4) / (100 + P3 * HC)
390  N = N + 1
400  IF N > 150 THEN GOTO 550
410  IF ABS (XN – XF) > EPS GOTO 350
420  TCL = 100 * XN - 273                                   : surface temperature of the clothing
430  --------------------------------HEAT LOSS COMPONENTS ----------------------------------
440  HL1 = 3.05 * .001 (5733-6.99 * MW-PA)                  : heat loss diff. through skin
450  IF MW > 58.15 THEN HL2 = .42 * (MW – 58.15)
     ELSE HL2 = 0!                                          : heat loss by sweating (comfort)
```

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```
460   HL3 = 1.7 * .00001 * M * (5867-PA)                    : latent respiration heat loss
470   HL4 = .0014 * M * (34 - TA)                           : dry respiration heat loss
480   HL5 = 3.96 * FCL * (XN^4 – (TRA/100)^4)               : heat loss by radiation
490   HL6 = FCL*HC*(TCL-TA)                                 : heat loss by convection
500   --------------------------------CALCULATE PMV AND PPD ----------------------------------
510   TS = .303 * EXP (- .036 * M) + .028                   : thermal sensation trans coeff
520   PMV = TS * (MW – HL1 – HL2 – HL3 – HL4 – HL5 –HL6)    : predicted mean vote
530   PPD = 100 – 95 * EXP (- .03353 * PMV ^ 4 - .2179 * PMV^   : predicted percentage dissat.
      2)
540   GOTO 570
550   PMV = 999999!
560   PPD = 100
570   PRINT:PRINT "OUTPUT"                                  : output
580   PRINT "          Predicted Mean Vote                  (PMV): "
      :PRINT USING "# # . #": PMV
590   PRINT "          Predicted Percent of Dissatisfied    (PPD): "
      :PRINT USING "# # # . #": PPD
600   PRINT: INPUT "NEXT RUN (Y/N)"; RS
610   IF (RS = "Y" OR RS = "y") THEN RUN
620   END
```

The following are some examples calculated by the listed program.

```
DATA ENTRY
  Clothing                                    (clo) ? 1.0
  Metabolic rate                              (met)? 1.2
  External work, normally around 0            (met) ? 0
  Air temperature                             ( C) ? 19.0
  Mean radiant temperature                    ( C) ? 18.0
  Relative air velocity                       (m/s) ? 0.1
ENTER EITHER RH OR WATER VAPOUR PRESSURE BUT NOT BOTH
  Relative humidity                           (%) ? 40
  Water vapour pressure                       (Pa) ?
```

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OUTPUT

Predicted Mean Vote (PMV): -0.7

Predicted Percent of Dissatisfied (PPD): 15.3

Table D.1 — Example output

Run no.	Air temperature °C	Mean radiant temperature °C	Relative air velocity m/s	RH %	Metabolic rate met	Clothing insulation clo	PMV	PPD
1	22,0	22,0	0,10	60	1,2	0,5	−0,75	17
2	27,0	27,0	0,10	60	1,2	0,5	0,77	17
3	27,0	27,0	0,30	60	1,2	0,5	0,44	9
4	23,5	25,5	0,10	60	1,2	0,5	−0,01	5
5	23,5	25,5	0,30	60	1,2	0,5	−0,55	11
6	19,0	19,0	0,10	40	1,2	1,0	−0,60	13
7	23,5	23,5	0,10	60	1,2	1,0	0,50	10
8	23,5	23,5	0,30	40	1,2	1,0	0,12	5
9	23,0	21,0	0,10	40	1,2	1,0	0,05	5
10	23,0	21,0	0,30	40	1,2	1,0	−0,16	6
11	22,0	22,0	0,10	60	1,6	0,5	0,05	5
12	27,0	27,0	0,10	60	1,6	0,5	1,17	34
13	27,0	27,0	0,30	60	1,6	0,5	0,95	24

Key

RH relative humidity

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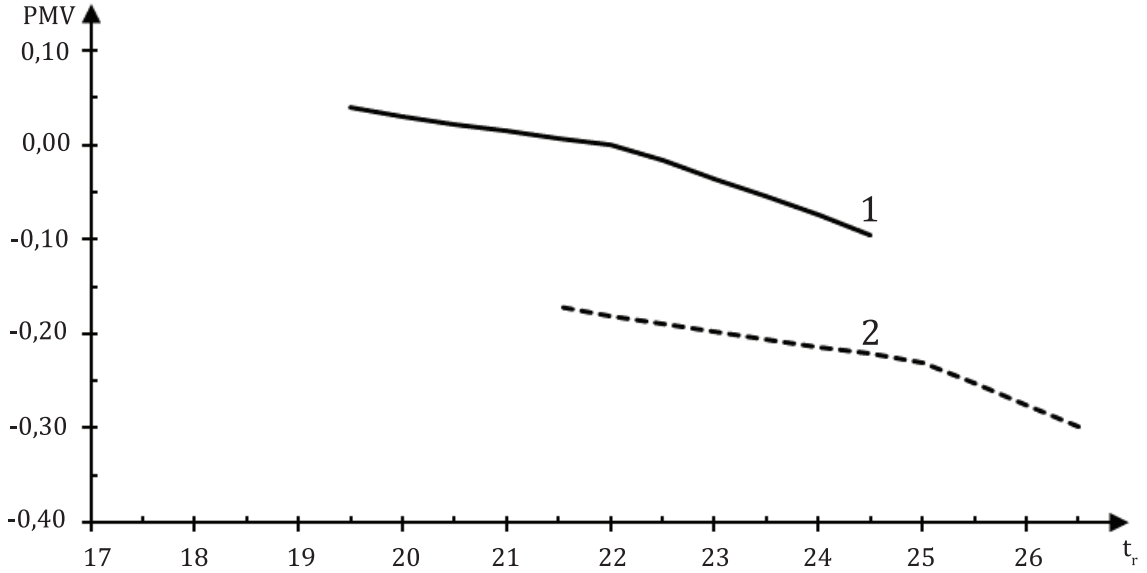
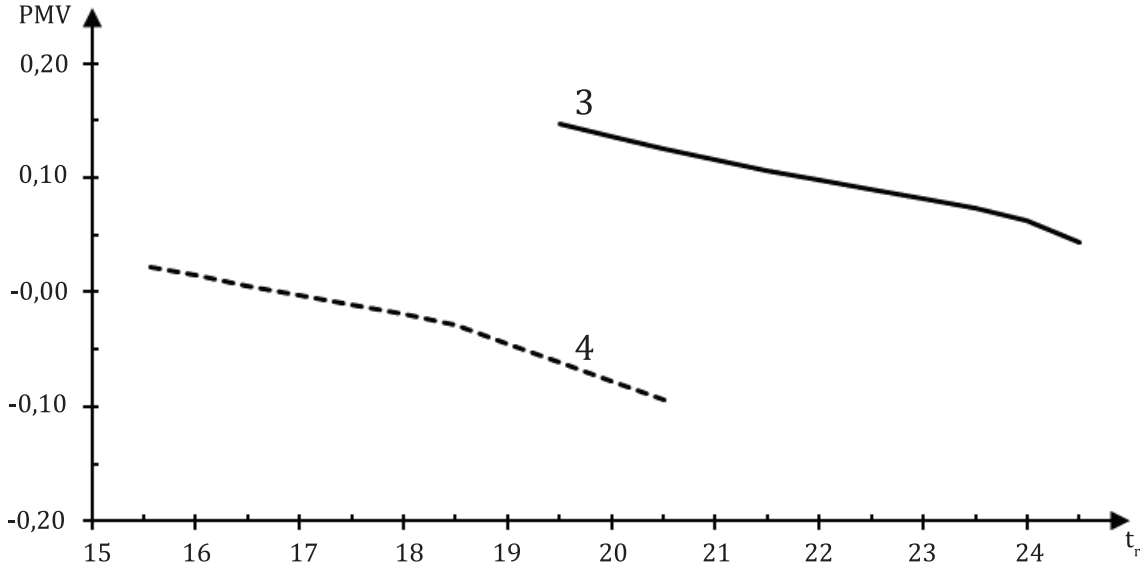
Annex E
(informative)

Graphics for determination of predicted mean vote (PMV)

The following graphs are for information purposes and easy estimation of PMV for many typical applications.





Key

t_r	mean radiant temperature, °C
PMV	predicted mean vote, 1
1	t_o =22,0 °C M=1,6 met

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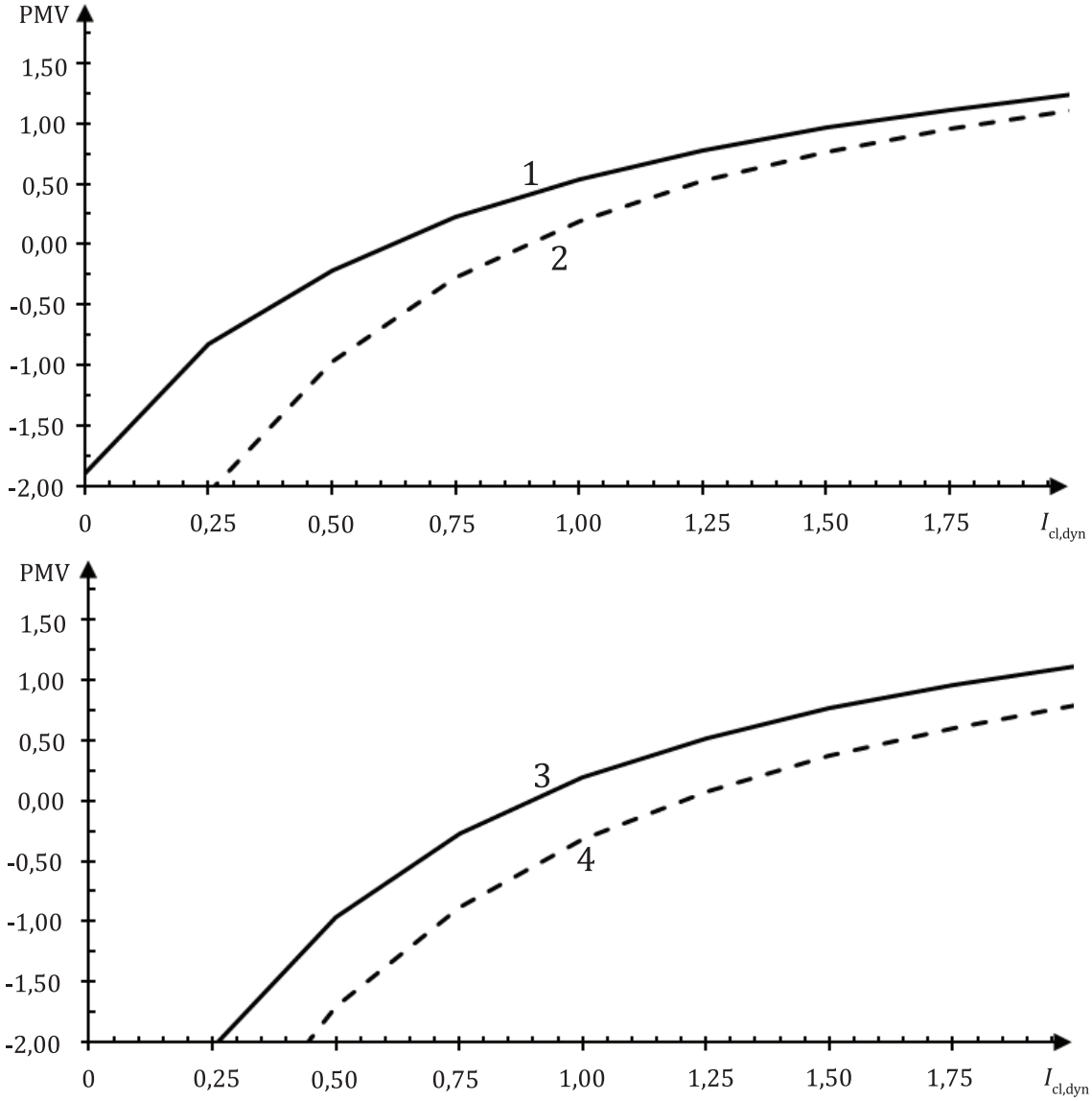
2 t_o =24,0 °C M=1,2 met

3 t_o =22,0 °C M=1,2 met

4 t_o =18,0 °C M=1,6 met

NOTE Figure E.1 shows PMV trends as a function of the mean radiant temperature t_r in summer ($I_{cl,dyn}$=0,50 clo) and winter ($I_{cl,dyn}$ =1,0 clo). Figure of the top (bottom) refers to summer (winter) conditions. Figure E.1 a) shows t_o=24 °C, M=1,2 met; Figure E.1 b) shows t_o =22 °C, M=1,6 met; Figure E.1 c) shows t_o =22 °C, M=1,2 met; Figure E.1 d) shows t_o =18 °C, M=1,6 met.

Figure E.1 — PMV trends as a function of the mean radiant temperature t_r in summer ($I_{cl,dyn}$=0,50 clo) and winter ($I_{cl,dyn}$ =1,0 clo)



Key

$I_{cl,dyn}$ dynamic clothing insulation, clo

PMV predicted mean vote, 1

1 var=0,1 m/s t_o =24,0 °C

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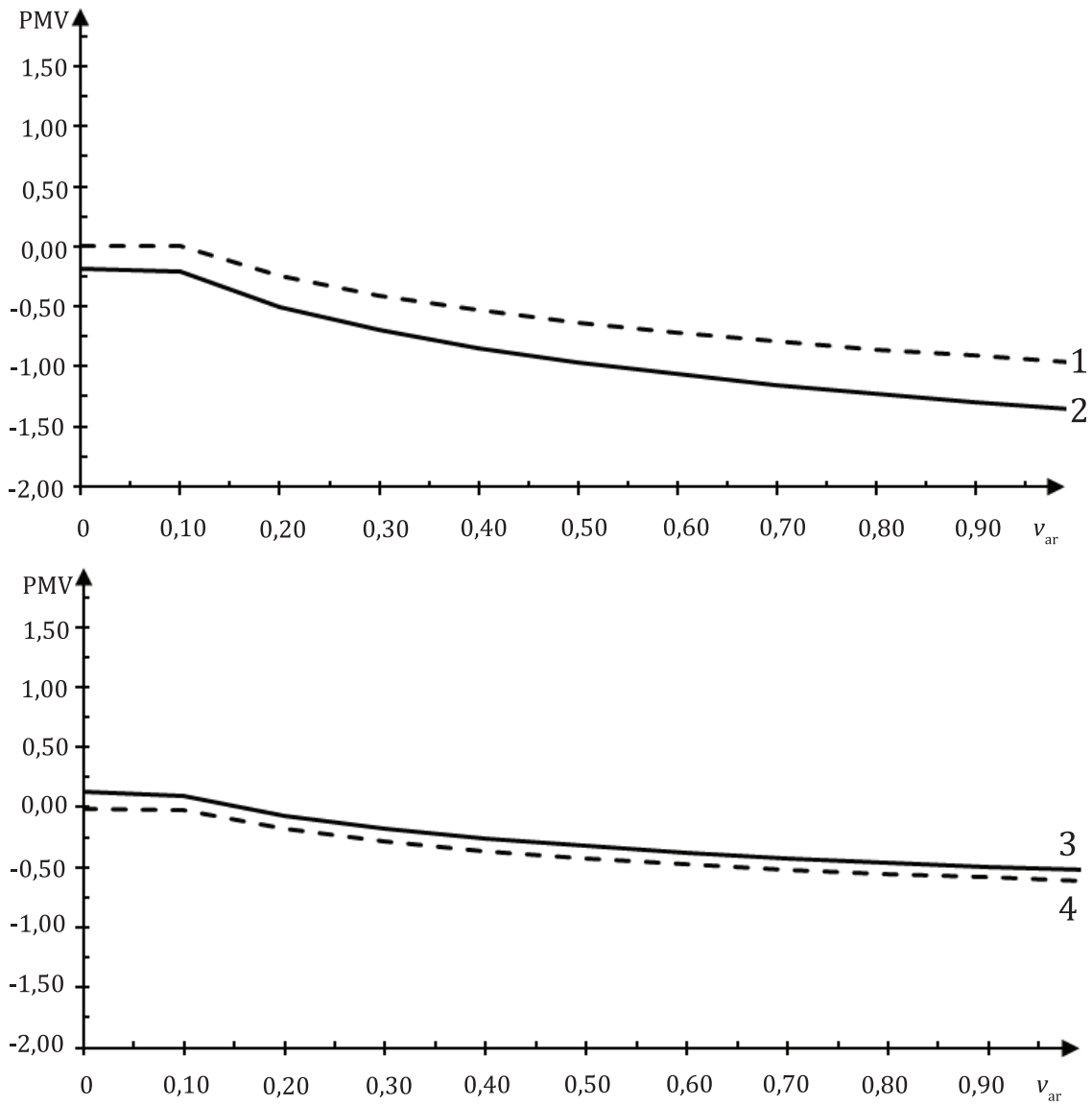
2 var=0,5 m/s t_o =24,0 °C

3 var=0,1 m/s t_o =22,0 °C

4 var=0,5 m/s t_o =22,0 °C

NOTE Figure E.2 shows PMV trends as a function of the dynamic clothing insulation $I_{cl,dyn}$ for two metabolic rates values: M=1,2 met (top) and M=1,6 met (bottom). Figure E.2 a) shows t_o=24 °C, v_{ar}=0,1 m/s; Figure E.2 b) shows t_o=24 °C, v_{ar}=0,5 m/s; Figure E.2 c) shows t_o=22 °C, v_{ar}=0,1 m/s; Figure E.2 d) shows d) t_o=22 °C, v_{ar}=0,1 m/s.

Figure E.2 — PMV trends as a function of the dynamic clothing insulation $I_{cl,dyn}$ for two metabolic rates values: M=1,2 met (top) and M=1,6 met (bottom)



Key

v_{ar} relative air velocity, m/s

PMV predicted mean vote, 1

1 t_o =22,0 °C M=1,6 met

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2 t_o =24,0 °C M=1,2 met

3 t_o =22,0 °C M=1,2 met

4 t_o =18,0 °C M=1,6 met

NOTE Figure E.3 shows PMV trends as a function of the relative air velocity v_{ar} in summer ($I_{cl,dyn}$=0,50 clo) and winter ($I_{cl,dyn}$ =1,0 clo). The figure of the top (bottom) refers to summer (winter) conditions. Figure E.3 a) shows t_o=24 °C, M=1,2 met; Figure E.3 b) shows t_o =22 °C, M=1,6 met; Figure E.3 c) shows t_o =22 °C, M=1,2 met; Figure E.3 d) shows t_o =18 °C, M=1,6 met.

Figure E.3 — PMV trends as a function of the relative air velocity v_{ar} in summer ($I_{cl,dyn}$=0,50 clo) and winter ($I_{cl,dyn}$ =1,0 clo)

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Annex F
(informative)

Humidity

Humidity can be expressed as relative or absolute humidity (see ISO 7726). The absolute humidity expressed as water vapour pressure in the air, which influences the evaporative heat loss from a person, and then the general thermal comfort, see Formula (1). At moderate temperatures (< 26 °C) and moderate activity levels (< 2 met) this influence is, however, rather limited. In moderate environments, the air humidity has only a modest impact on the thermal sensation. Typically, a 10 % higher relative humidity is felt to be as warm as a 0,3 °C rise in the operative temperature. For higher temperatures and activities, the influence is greater. Under transient conditions, the humidity can also have a significant influence.

If humidity limits are based on the maintenance of acceptable thermal conditions based solely on comfort considerations, including thermal sensation, skin wetness, skin dryness and eye irritation, a wide range of humidity is acceptable.

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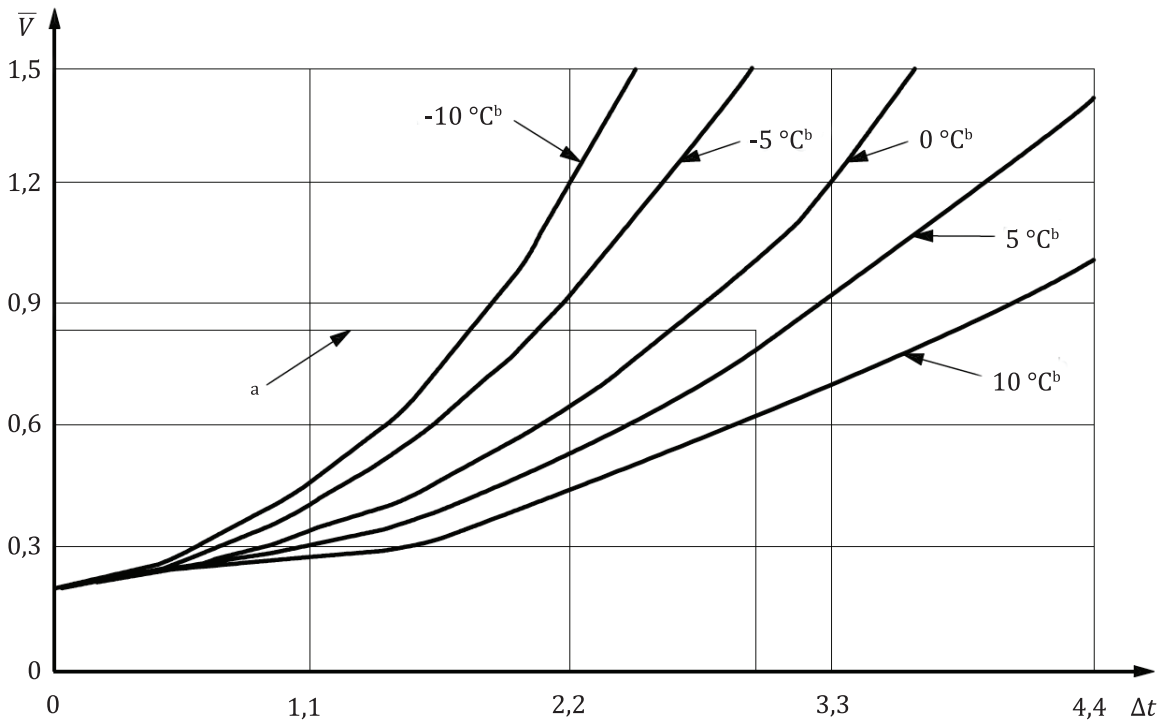
Annex G
(informative)

Air velocity

The air velocity in a space influences the convective heat exchange between a person and the environment. This influences the general thermal comfort of the body (heat loss) expressed by the PMV-PPD index (see Clauses 4 and 5) and the local thermal discomfort due to draught (Clause 6). There is no minimum air velocity that is necessary for thermal comfort. However, increased air velocity can be used to offset the warmth sensation caused by increased temperature.

Often, the air velocity is increased by opening of windows or use of fans to adapt to warmer environments. Under summer conditions, the temperature can be increased above the level allowed for comfort if a means is provided to also elevate the air velocity. The amount by which the temperature may be increased is shown in Figure G.1. The combinations of air velocity and temperature defined by the lines in the figure result in the same heat transfer from the skin. The reference point for these curves is 26 °C and 0,20 m/s of air velocity. The benefits that can be gained by increasing air velocity depend on clothing, activity and the difference between the surface temperature of the clothing or skin and the air temperature. Figure G.1 shows the air velocity that is required for typical summer clothing (0,5 clo) and sedentary activities (1,2 met) that correspond to summer comfort.



For light primarily sedentary activity, Δt should be < 3 °C and $\overline{v}$ < 0,82 m/s.

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Key

Δt temperature rise above 26 °C

$\bar{v}$ mean air velocity, m/s

a Limits for light, primarily sedentary, activity.

b ($\bar{t}_r - t_a$), °C (t_a, air temperature, °C; $\bar{t}_r$, mean radiant temperature, °C).

Figure G.1 — Air velocity required to offset increased temperature

Figure G.1 applies to an increase of temperature above 26 °C, with both t_r and t_a increasing equally. When the mean radiant temperature is low and the air temperature is high, elevated air velocity is less effective at increasing heat loss. Conversely, elevated air velocity is more effective at increasing heat loss when the mean radiant temperature is high and the air temperature is low. Therefore, the curve in Figure G.1, that corresponds to the relative difference between air temperature and mean radiant temperature, must be used. Large individual differences exist between people with regard to the preferred air velocity. Therefore, the elevated air velocity must be under the direct control of the affected occupants and adjustable in steps no greater than 0,15 m/s.

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